QUBO Formulation: HP Lattice Protein Conformation Selection

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Problem Statement. Given a hydrophobic-polar protein sequence and a curated set of M candidate lattice conformations, each associated with a precomputed contact energy E_j, the task is to select exactly one conformation that minimizes the total contact energy. The contact energy of a conformation is determined by the number of nonconsecutive hydrophobic-hydrophobic contacts it forms, with each contact contributing -1 to the total energy.

Decision Variables. For each candidate conformation $j \in \{0, 1, \ldots, M-1\}$, we introduce a binary decision variable $x_j \in \{0, 1\}$. The variable x_j equals 1 if conformation j is selected, and 0 otherwise. The vector of decision variables is denoted $\mathbf{x} = (x_0, x_1, \ldots, x_{M-1})^\top$.

QUBO Derivation. We construct the Quadratic Unconstrained Binary Optimization (QUBO) formulation by converting the constrained minimization problem into an unconstrained energy minimization problem via a penalty method.

Step 1. The original objective is to minimize the total contact energy over the binary variables:

$$\min_{\mathbf{x}\in\{0,1\}^M} \sum_{j=0}^{M-1} E_j x_j. \tag{1}$$

Step 2. The selection requirement is enforced by the equality constraint that exactly one conformation must be chosen:

$$\sum_{j=0}^{M-1} x_j = 1. \tag{2}$$

Step 3. We convert the constraint into a quadratic penalty term with weight $\lambda > 0$:

$$P(\mathbf{x}) = \lambda \left(\sum_{j=0}^{M-1} x_j - 1 \right)^2 \tag{3}$$

$$= \lambda \left(\sum_{j=0}^{M-1} x_j^2 + \sum_{j \neq k} x_j x_k - 2 \sum_{j=0}^{M-1} x_j + 1 \right) \tag{4}$$

$$= \lambda \left(\sum_{j=0}^{M-1} x_j + \sum_{j \neq k} x_j x_k - 2 \sum_{j=0}^{M-1} x_j + 1 \right) \tag{5}$$

$$= \lambda \left(\sum_{j \neq k} x_j x_k - \sum_{j=0}^{M-1} x_j + 1 \right), \tag{6}$$

where the second equality follows from expanding the square, and the third equality uses the idempotent property $x_j^2 = x_j$ for binary variables. The cross-term sum $\sum_{j \neq k} x_j x_k$ can be rewritten over unique pairs as $2 \sum_{0 \leq j < k \leq M-1} x_j x_k$.

Step 4. Combining the objective and the penalty term yields the QUBO Hamiltonian $H(\mathbf{x})$:

$$H(\mathbf{x}) = \sum_{j=0}^{M-1} E_j x_j + \lambda \left(2 \sum_{0 \leq j < k \leq M-1} x_j x_k - \sum_{j=0}^{M-1} x_j + 1 \right) \tag{7}$$

$$= \sum_{j=0}^{M-1} (E_j - \lambda) x_j + 2\lambda \sum_{0 \leq j < k \leq M-1} x_j x_k + \lambda. \tag{8}$$

Step 5. Reading off the coefficients from the standard QUBO form $H(\mathbf{x}) = \sum_{j \leq k} Q_{jk} x_j x_k + c$, we obtain the general closed-form expressions for the upper-triangular Q matrix entries and the constant offset:

$$Q_{jj} = E_j - \lambda, \quad \forall j \in \{0, \ldots, M-1\}, \tag{9}$$

$$Q_{jk} = 2\lambda, \quad \forall 0 \leq j < k \leq M-1, \tag{10}$$

$$c = \lambda. \tag{11}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any constraint violation is energetically unfavorable compared to the best feasible solution. The minimum penalty incurred by an infeasible assignment (where $\sum_{j=0}^{M-1} x_j \neq 1$) is λ, occurring when the sum equals 0 or 2. The maximum possible reduction in the objective energy achievable by switching from a feasible state to an infeasible state is bounded by $\max_j |E_j|$. Therefore, setting

$$\lambda > \max_{j \in \{0, \ldots, M-1\}} |E_j| \tag{12}$$

guarantees that the penalty cost strictly dominates any potential objective gain from violating the selection constraint.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^M$ recovers the optimal conformation because feasible solutions satisfy $\sum_{j=0}^{M-1} x_j = 1$ and thus incur zero penalty, reducing $H(\mathbf{x})$ exactly to the original objective $\sum_{j=0}^{M-1} E_j x_j$. Infeasible solutions incur a penalty of at least λ. Since λ exceeds the maximum possible objective value magnitude, any infeasible assignment has strictly higher energy than the optimal feasible assignment. Consequently, the global minimum of $H(\mathbf{x})$ must be feasible and correspond to the conformation with the lowest contact energy.

Illustrative Example. Consider a small instance with $M = 3$ candidate conformations and precomputed energies E_0, E_1, and E_2. Substituting these parameters into the general formulas yields the concrete Hamiltonian:

$$H(\mathbf{x}) = E_0 x_0 + E_1 x_1 + E_2 x_2 + 2\lambda(x_0 x_1 + x_0 x_2 + x_1 x_2) - \lambda(x_0 + x_1 + x_2) + \lambda. \tag{13}$$

The corresponding Q matrix entries and offset are:

$$Q_{00} = E_0 - \lambda, \quad Q_{11} = E_1 - \lambda, \quad Q_{22} = E_2 - \lambda, \tag{14}$$

$$Q_{01} = Q_{02} = Q_{12} = 2\lambda, \quad c = \lambda. \tag{15}$$

With $\lambda > \max(|E_0|, |E_1|, |E_2|)$, the global minimum of $H(\mathbf{x})$ occurs at the binary vector $\mathbf{x}^*$ where $x_j^* = 1$ for the index j that minimizes E_j, and $x_k^* = 0$ for all $k \neq j$. The minimum energy value is exactly $\min(E_0, E_1, E_2)$, correctly identifying the optimal conformation and its contact energy.